

PROJECT REPORT

Exploratory Data Analysis, Data Visualization, and a Product Recommendation System on the Amazon Sales Dataset

A Data Analytics Project

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Tools: Python (Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn) | Google Colab

August 2026

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1. Abstract

This project performs an end-to-end analysis of a publicly listed Amazon India product dataset comprising 1,465 product-review records across 211 fine-grained categories. The work spans three stages: (1) data cleaning and exploratory data analysis (EDA) to understand pricing, discounting, and rating patterns; (2) data visualization to surface trends across categories and price bands; and (3) the design and implementation of a recommendation system using three complementary techniques — popularity-based ranking, content-based filtering (TF-IDF and cosine similarity), and an item-based collaborative filtering demonstration built from embedded reviewer identifiers. The analysis was implemented in Python and packaged as a self-contained Google Colab notebook.

2. Introduction

E-commerce platforms such as Amazon generate large volumes of structured and unstructured data through product listings and customer reviews. Understanding this data helps sellers price competitively, helps platforms surface relevant products, and helps shoppers make faster decisions. This project uses a dataset of Amazon product listings — including pricing, discounting, aggregate star ratings, and free-text reviews — to practice the full analytics workflow: cleaning messy real-world data, exploring it visually, and building a working recommendation system on top of it.

2.1 Objectives

- Clean and standardize a real-world, messily formatted e-commerce dataset.
- Perform exploratory data analysis to understand category distribution, pricing, discounting, and rating behaviour.
- Visualize key patterns to make findings interpretable to a non-technical audience.
- Design and implement a multi-method recommendation system (popularity-based, content-based, and collaborative-filtering-style).
- Deliver a reproducible, well-documented Google Colab notebook.

3. Dataset Description

The dataset contains 1,465 rows and 16 columns, describing 1,351 unique products spread across 211 sub-categories (9 top-level categories). Each row also embeds the identifiers, names, and text of up to eight customer reviews associated with that product.

Column	Description
product_id / product_name	Unique identifier and title of the product
category	Pipe-separated category hierarchy (e.g. Electronics ... USBCables)
discounted_price / actual_price	Selling price and list price, in ₹
discount_percentage	Percentage discount off the list price

Column	Description
rating / rating_count	Aggregate star rating (1–5) and number of ratings
about_product	Bullet-point product description
user_id / user_name / review_id	Reviewer identifiers and names (comma-separated, multiple per product)
review_title / review_content	Short and long review text
img_link / product_link	Product image and Amazon listing URL

Note: despite several columns referencing “price,” the dataset contains no transaction or unit-sold data — it is a snapshot of product listings and their aggregate ratings, not a sales ledger. Throughout this report, `rating_count` is used as a proxy for popularity.

4. Methodology

4.1 Data Cleaning

The raw fields for price, discount, and rating count were stored as text (e.g. “₹1,099”, “64%”, “24,269”) and required conversion to numeric types by stripping currency symbols, thousands separators, and percentage signs. One row contained a malformed rating value (“|” instead of a number) and was excluded after cleaning, along with two rows missing `rating_count`, leaving 1,462 usable rows. The pipe-separated category string was split to derive a `main_category` field for higher-level grouping.

4.2 Exploratory Data Analysis

EDA covered the distribution of ratings, the spread of products across categories, price and discount statistics, and the relationship between discounting, price, and rating. Descriptive statistics and correlation analysis were used to quantify these relationships before visualizing them.

4.3 Data Visualization

Eight visualizations were produced using Matplotlib and Seaborn: a rating histogram, a top-10 category bar chart, a log-scaled price distribution, a discount-vs-rating scatter plot, average rating by category, a price-vs-rating bubble chart sized by review volume, a correlation heatmap, and a word cloud of review text. The most illustrative charts are reproduced in Section 6.

4.4 Recommendation System Design

Three recommendation approaches were implemented to cover complementary use cases:

- Popularity-based: an IMDB-style weighted rating (bayesian average) that balances a product's star rating against how many people rated it, preventing a 5.0 rating from 3 reviewers from outranking a 4.6 rating from 50,000 reviewers. Supports overall or per-category “top products” lists.
- Content-based filtering: product name, category, and description text are combined and vectorized with TF-IDF; cosine similarity between vectors surfaces “more like this” recommendations, which works even for products with no reviews yet.

- Item-based collaborative filtering (demonstration): the dataset embeds multiple reviewer IDs per product row. These were exploded into a user–item interaction table, and item-item cosine similarity was computed from shared reviewers (co-review overlap), approximating collaborative filtering in the absence of individual per-user star ratings.

5. Tools and Technologies

Category	Tools Used
Language / Environment	Python 3, Google Colab
Data handling	Pandas, NumPy
Visualization	Matplotlib, Seaborn, WordCloud
Machine Learning	Scikit-learn (TF-IDF Vectorizer, cosine similarity)
Sparse matrix / CF	SciPy (csr_matrix)

6. Results and Key Findings

6.1 Dataset Summary Statistics

Metric	Value
Rows analysed (after cleaning)	1,462
Unique products	1,348
Top-level categories	9
Average rating	4.10 / 5
Median rating	4.10 / 5
Average discount	47.7%
Average discounted price	₹3,129.98
Average actual (list) price	₹5,453.09
Highest rating_count for a single product	426,973
Largest category	Electronics (526 products)

6.2 Rating Distribution

Ratings are heavily concentrated between 3.8 and 4.5, with very few products rated below 3.0 or at a perfect 5.0. This right-skewed, tightly-clustered pattern is typical of curated e-commerce catalogues, where poorly rated products tend to be delisted or receive few reviews.

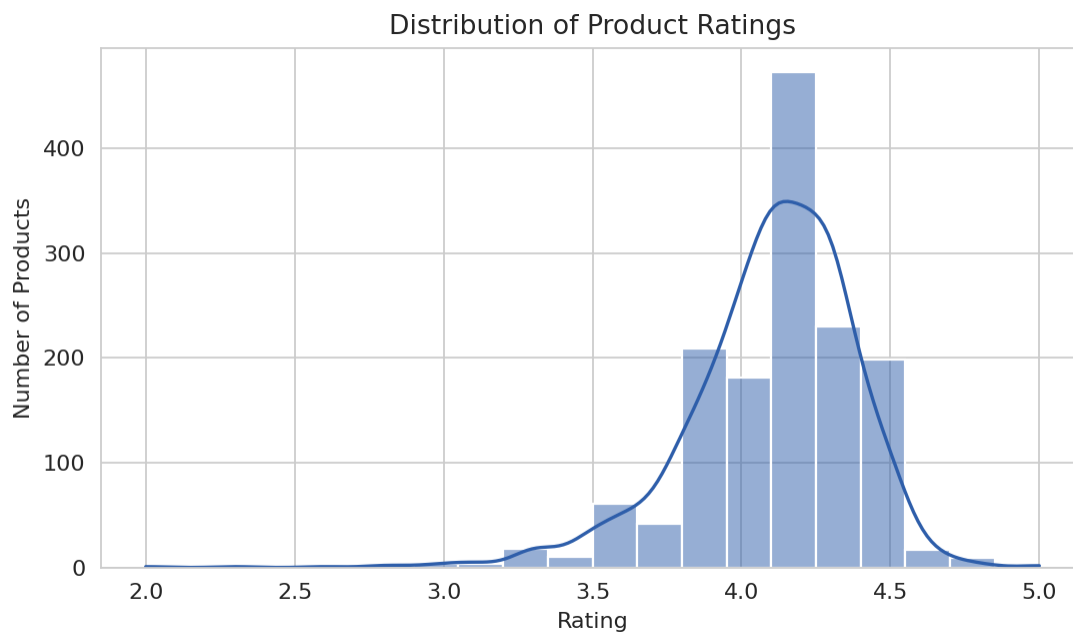


Figure 1: Distribution of product ratings across the catalogue.

6.3 Category Composition

Electronics dominates the catalogue with 526 products, followed by Computers & Accessories and Home & Kitchen. This concentration reflects the source listing mix rather than platform-wide category share, and is important context when interpreting category-level averages elsewhere in this report.

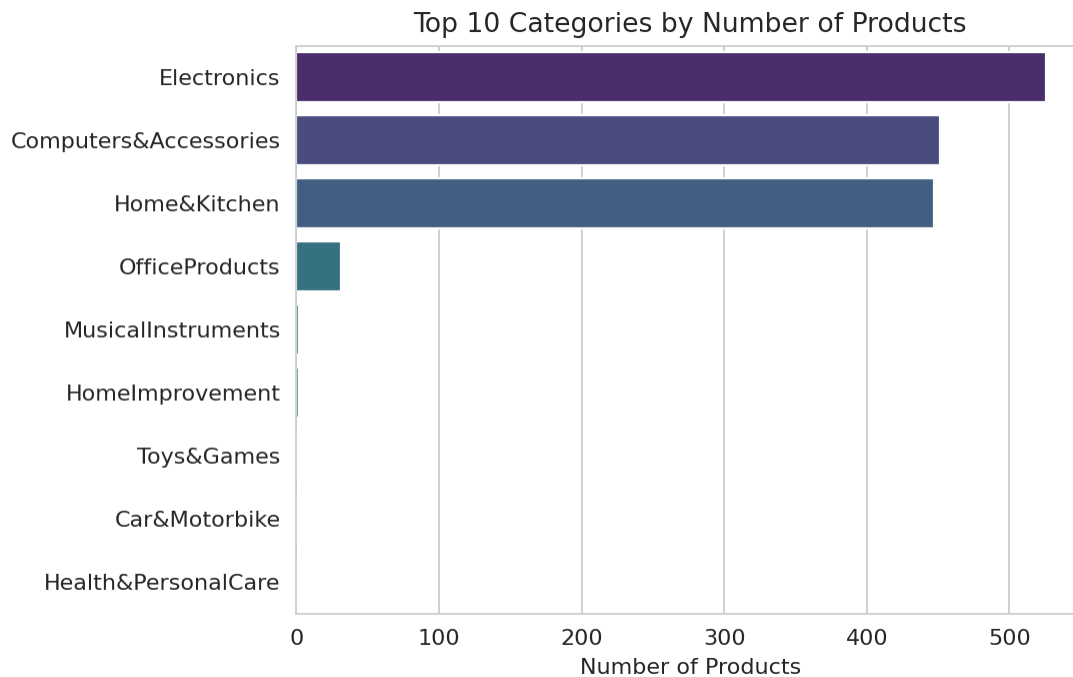


Figure 2: Top 10 categories by number of products.

6.4 Discount Percentage vs Rating

The correlation between discount percentage and rating is weakly negative ($r = -0.156$), suggesting heavily discounted products are, on average, rated very slightly lower — though the effect is small and the scatter shows wide variance at every discount level. Price itself has a weak positive correlation with rating ($r = 0.121$), meaning higher-priced items skew marginally higher-rated, again a weak effect.

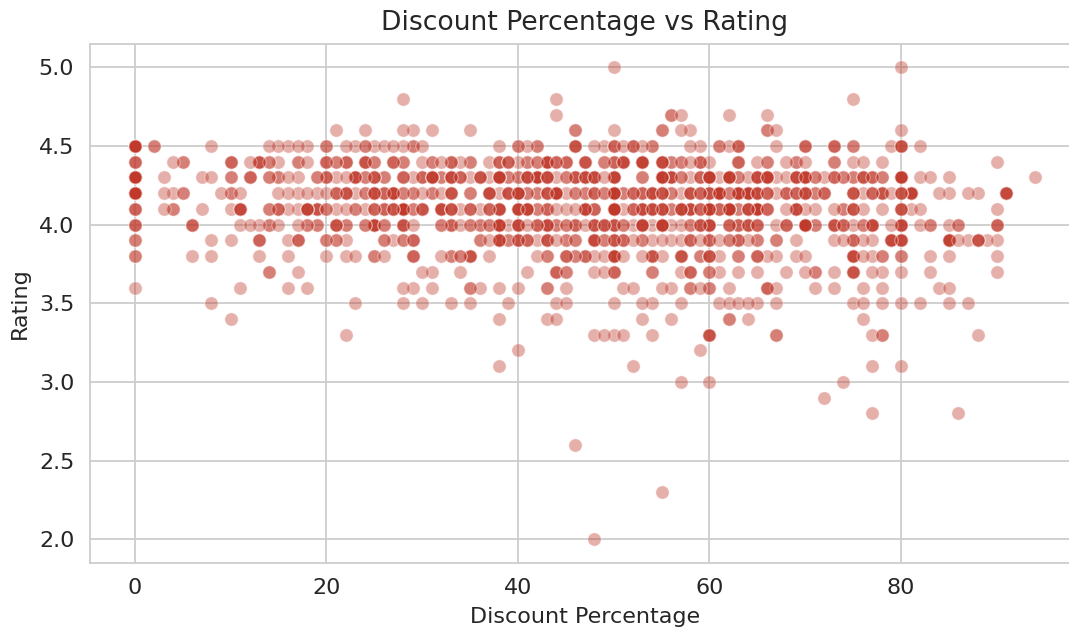


Figure 3: Discount percentage plotted against product rating.

6.5 Correlation Overview

The heatmap below summarizes pairwise correlations between price, discount, rating, and rating_count. No strong linear relationships were found among these numeric fields, indicating that rating quality is driven mainly by factors outside this dataset (e.g. product category, brand, or the review text itself) rather than by price or discount alone.

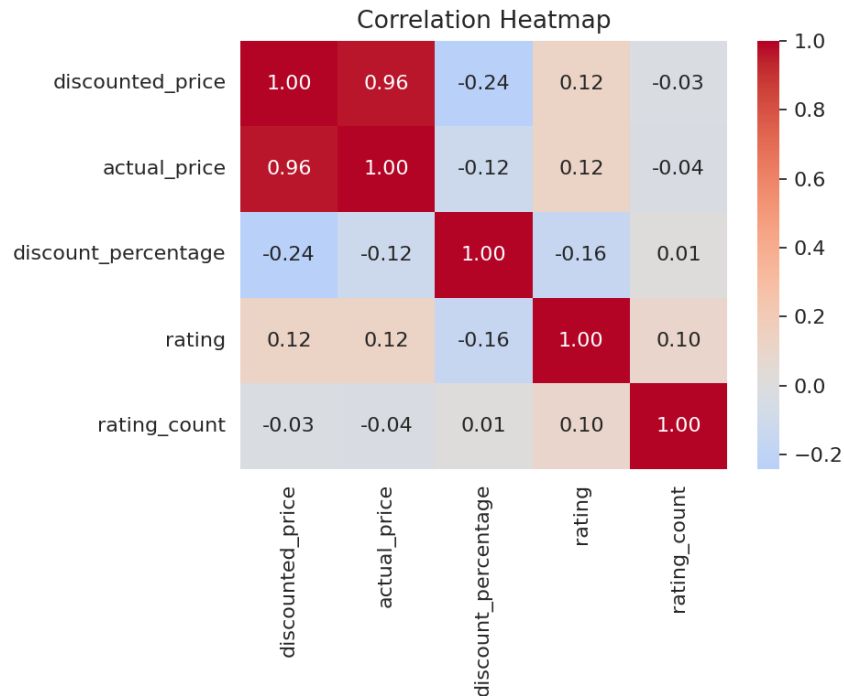


Figure 4: Correlation heatmap of numeric fields.

6.6 Recommendation System Outcomes

All three recommenders were validated on the cleaned dataset:

- The popularity-based ranker correctly surfaces products with both high ratings and high review volume (e.g. a water heater faucet tap and a SanDisk memory card topped the overall weighted-score ranking), avoiding the bias of unweighted average rating.
- The content-based recommender, queried with the keyword “headphone,” returned topically and semantically related cable and charger accessories with descending cosine similarity scores, confirming that the TF-IDF vectors meaningfully capture product similarity from text.
- The collaborative-filtering demonstration successfully built a user–item interaction matrix from the embedded reviewer IDs and produced item-item similarity scores based on shared reviewers, illustrating the mechanics of CF even though this dataset lacks true per-user star ratings.

7. Challenges and Limitations

- Price, discount, and rating fields required manual text cleaning due to inconsistent formatting (currency symbols, thousands separators, percentage signs, and one corrupted rating value).
- The dataset provides only an aggregate product rating rather than individual user ratings, limiting collaborative filtering to a co-review-overlap approximation rather than true rating-based CF.
- No transaction or sales-volume data is present, so “sales performance” could only be approximated via `rating_count` as a popularity proxy.
- Category imbalance (Electronics accounts for over a third of listings) means category-level averages should be read with sample size in mind.

8. Conclusion and Future Scope

This project demonstrates a complete data analytics pipeline — from raw, inconsistently formatted data to cleaned datasets, visual insights, and a working multi-method recommendation system. The findings show a catalogue of generally well-rated products with heavy discounting and only weak relationships between price, discount, and rating. The three recommenders together cover the most common recommendation scenarios: ranking by overall popularity, finding similar items by content, and approximating collaborative filtering from reviewer overlap.

Future work could extend this project by: (1) sourcing true per-user, per-product ratings to enable proper collaborative filtering and matrix factorization (e.g. SVD); (2) building a hybrid recommender that blends content similarity with popularity score and category filters; (3) adding sentiment analysis on `review_content` to weight recommendations by review sentiment, not just star rating; and (4) deploying the recommender behind a simple web interface for interactive use.

9. References

- Dataset: Amazon Sales Dataset (Kaggle) — product listings, prices, ratings, and reviews from [amazon.in](https://www.kaggle.com/datasets/ashishpatel26/amazon-sales).
- Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, SciPy — official documentation.
- Google Colaboratory — colab.research.google.com.